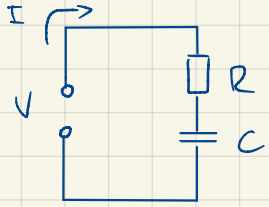


2) RC-circuit, impedance



$$V(t) = R \cdot I(t) + \frac{Q(t)}{C} \quad (*)$$

$$I(t) = \dot{Q}(t)$$

$$\hat{I}(\omega) \stackrel{!}{=} i\omega \hat{Q}(\omega)$$

impedance: $\hat{V}(\omega) = Z(\omega) \hat{I}(\omega)$

from (*):
(linearity of FT)

$$\hat{V}(\omega) = R \cdot \hat{I}(\omega) + \frac{\hat{Q}(\omega)}{C} = R \cdot \hat{I}(\omega) + \frac{\hat{I}(\omega)}{i\omega C}$$

so, $Z(\omega) = R + \frac{1}{i\omega C}$

$\omega \rightarrow \infty$: $Z(\omega) \rightarrow R$: high frequency : C doesn't matter

$\omega \rightarrow 0$: $Z(\omega) \rightarrow \infty$: for DC, C is a wall

Meaning of impedance:

harmonic signal: $I(t) = I_0 \cos(\omega t) \rightarrow Q(t) = \frac{I_0}{\omega} \sin(\omega t)$

from (*): $U(t) = I_0 R \cos(\omega t) + \frac{I_0}{\omega C} \sin(\omega t)$

$$= I_0 \underbrace{\sqrt{R^2 + \frac{1}{(\omega C)^2}}}_{|Z(\omega)|} \cos(\omega t - \varphi) \quad \leftarrow \tan \varphi = \frac{1/\omega C}{R} = \frac{1}{\omega R C}$$

$\uparrow$
 ! Arg Z(ω)

because

$$\cos(\omega t - \varphi) = \cos(\omega t) \cos \varphi + \sin(\omega t) \sin \varphi =$$

assume
 $-\pi/2 \leq \varphi \leq \pi/2$

$$= \cos(\omega t) \frac{1}{\sqrt{1 + \tan^2 \varphi}} + \sin(\omega t) \frac{\tan \varphi}{\sqrt{1 + \tan^2 \varphi}}$$

$$= \cos(\omega t) \frac{1}{\sqrt{1 + \frac{1}{(\omega R C)^2}}} + \sin(\omega t) \frac{1/\omega R C}{\sqrt{1 + \frac{1}{(\omega R C)^2}}}$$

so

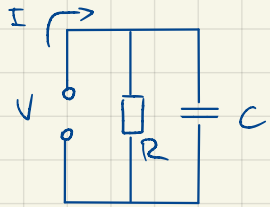
$$I_0 R \sqrt{1 + \frac{1}{(\omega R C)^2}} \cos(\omega t - \varphi) = I_0 R \cos(\omega t) + \frac{I_0}{\omega C} \sin(\omega t) \quad \checkmark$$

/ If $\pi/2 < \varphi \leq \pi$ or $-\pi \leq \varphi < -\pi/2$ there is an extra overall sign. /

so, $U(t) = \pm I_0 \cdot |Z(\omega)| \cdot \cos(\omega t - \varphi) \quad \varphi = \text{Arg } Z(\omega)$

series circuit: $Z(\omega) = Z_R(\omega) + Z_C(\omega) = R + \frac{1}{i\omega C}$

Parallel circuit:



$$V(t) = V_R(t) = V_C(t)$$

$$I(t) = I_R(t) + I_C(t)$$

$$\hat{V}(\omega) = \hat{V}_R(\omega) = R \cdot \hat{I}_R(\omega) = Z_R(\omega) \cdot \hat{I}_R(\omega)$$

$$\hat{V}(\omega) = \hat{V}_C(\omega) = \frac{1}{i\omega C} \hat{I}_C(\omega) = Z_C(\omega) \hat{I}_C(\omega)$$

$$\hat{I}(\omega) = \hat{I}_R(\omega) + \hat{I}_C(\omega) = \frac{\hat{V}(\omega)}{Z_R(\omega)} + \frac{\hat{V}(\omega)}{Z_C(\omega)}$$

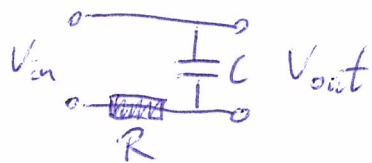
So,

$$\frac{\hat{I}(\omega)}{\hat{V}(\omega)} = \frac{1}{Z(\omega)} = \frac{1}{Z_R(\omega)} + \frac{1}{Z_C(\omega)} \quad \checkmark$$

LPF box signal

Last time

$$\hat{V}_{out}(\omega) = \frac{1}{1 + i\omega RC} \hat{V}_{in}(\omega)$$



$$\hat{f}_{LPF}(\omega)$$

response function

Inverse FT of $\hat{f}_{LPF}(\omega) = \frac{1}{1 + i\omega RC}$

we know $\mathcal{F}(\Theta(t) e^{-at}) = \frac{1}{\sqrt{2\pi}} \frac{1}{a + i\omega}$

$$\hat{f}_{LPF}(\omega) = \frac{\sqrt{2\pi}}{\sqrt{2\pi}} \frac{\frac{1}{RC}}{\frac{1}{RC} + i\omega}$$

$$f_{LPF}(t) = \frac{\sqrt{2\pi}}{RC} \Theta(t) e^{-\frac{t}{RC}}$$

Let the input signal be $V_{in}(t) = \begin{cases} V_0 & \text{if } 0 < t < T \\ 0 & \text{else} \end{cases}$

What is $V_{out}(t)$?

Use convolution then:

$$\mathcal{F}^{-1}[\hat{f}(\omega) \hat{g}(\omega)](t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt' f(t-t') g(t')$$

$$V_{out}(t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} dt' V_{in}(t-t') f_{LPF}(t')$$

where is the integrand nonzero?

$$\begin{aligned} & 0 < t-t' < T & t' > 0 \\ & -t < -t' < T-t \\ & t > t' > t-T \end{aligned}$$

2 cases:

$\begin{cases}$

if $t-T > 0$

if $t-T < 0$

$$\begin{aligned} & \int_0^t dt' \\ & \int_{t-T}^t dt' \end{aligned}$$

$$(1) \quad t - T > 0 \quad t > T$$

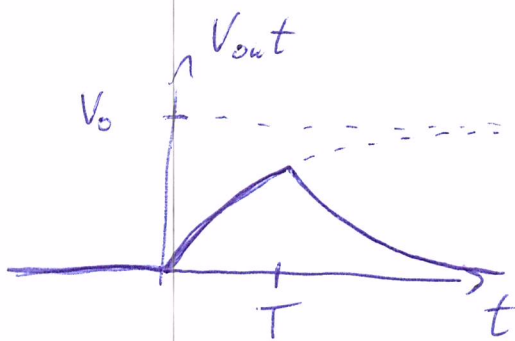
$$V_{out}(t) = \frac{V_0}{RC} \int_{t-T}^t dt' e^{-\frac{t'}{RC}} = -V_0 \left[e^{-\frac{t'}{RC}} \right]_{t-T}^t =$$

$$= -V_0 \left(e^{-\frac{t}{RC}} - e^{-\frac{t-T}{RC}} \right) = e^{-\frac{t-T}{RC}} \left(1 - e^{-\frac{T}{RC}} \right) \cdot V_0$$

$$(2) \quad t - T < 0 \quad 0 < t < T$$

$$V_{out}(t) = \frac{V_0}{RC} \int_0^t dt' e^{-\frac{t'}{RC}} = -V_0 \left[e^{-\frac{t'}{RC}} \right]_0^t = -V_0 \left(e^{-\frac{t}{RC}} - 1 \right) =$$

$$= V_0 \left(1 - e^{-\frac{t}{RC}} \right)$$

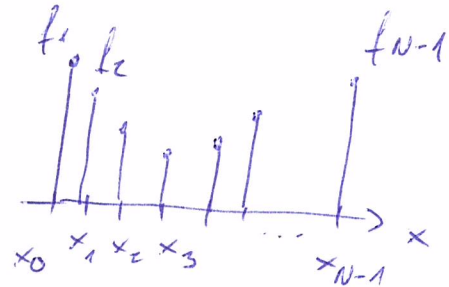


$$V_{out}(t) = \begin{cases} 0 & \text{if } t < 0 \\ V_0(1 - e^{-\frac{t}{RC}}) & \text{if } 0 < t < T \\ V_0(1 - e^{-\frac{T}{RC}})e^{-\frac{t-T}{RC}} & \text{if } T < t \end{cases}$$

DFT

Discretized signal $f_k = f(x_k)$ $k=0, \dots, N-1$

periodic/finite



discrete FT: also discretized $n=0, \dots, N-1$

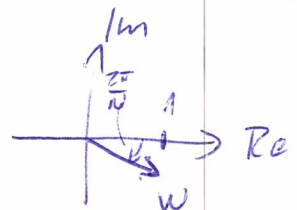
$$\hat{f}_n = \sum_{k=0}^{N-1} f_k e^{-2\pi i \frac{nk}{N}} = \sum_{k=0}^{N-1} f_k w^{nk}$$

where $w = e^{-\frac{2\pi i}{N}}$
N-th root of unity

Can also write as matrix product

$$\hat{f} = F f$$

$$\hat{f}_n = \sum_k F_{nk} f_k \quad \text{where } F_{nk} = w^{nk}$$



Inverse with matrix inverse

$$f = F^{-1} \hat{f}$$

Claim: $F^{-1} = \frac{1}{N} F^*$ $\leftarrow$ complex conjugate

Test for $N=3$

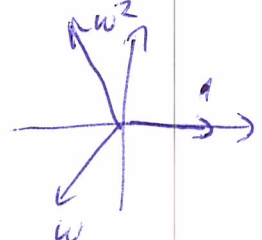
$$F^* = \begin{pmatrix} 1 & 1 & 1 \\ 1 & w^{-1} & w^{-2} \\ 1 & w^2 & w^{-1} \end{pmatrix}$$

e.g. for $N=3$
 $w = e^{-\frac{2\pi i}{3}}$ $w^3 = 1$

$$F = \begin{pmatrix} 1 & w & w^2 \\ w & w^2 & 1 \\ 1 & w^2 & w \end{pmatrix}$$

$$F F^* = \dots = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} = 3 \cdot \mathbf{1}$$

using $1 + w + w^2 = 0$



General proof:

$$(FF^*)_{nm} = \sum_{k=0}^{N-1} F_{nk} F_{km}^* = \sum_{k=0}^{N-1} w^{nk} w^{-km} = \sum_{k=0}^{N-1} w^{k(n-m)} = N \delta_{nm}$$

using that for any N 'th root of unity $\sum_{k=0}^{N-1} w^{ke} = \begin{cases} N & \text{if } e=0 \\ 0 & \text{otherwise} \end{cases}$

because this is a geometric sum

$$\sum_{k=0}^{N-1} (w^e)^k = \frac{1-(w^e)^N}{1-w^e} = \frac{1-(w^N)^e}{1-w^e} = \frac{0}{1-w^e} = 0 \text{ if } e \neq N$$

$$\text{if } e = N \quad \sum_{k=0}^{N-1} (w^N)^k = \sum_{k=0}^{N-1} 1 = N$$

So the inverse DFT is

$$f_k = (F^{-1} \hat{f})_k = \frac{1}{N} (F^* \hat{f})_k = \frac{1}{N} \sum_n F_{kn}^* \hat{f}_n = \frac{1}{N} \sum_n \hat{f}_n w^{nk}$$

Note F is symmetric $F = F^T$

$\Rightarrow F^*$ is symmetric $F^* = F^{*T} = F^+$

$$F^{-1} = \frac{1}{N} F^+ \quad \text{almost unitary (recall } U^{-1} = U^+)$$

sometimes we use the unitary version of DFT

$$\text{with } U = \frac{1}{\sqrt{N}} F, \quad U^{-1} = \frac{1}{\sqrt{N}} F^+$$